Study B — do the three contraceptive methods differ in community composition?

The question: do the vaginal communities of people on the pill, the injectable, and the ring differ from each other?

The plan, built on Tuesday's ordination module:

1. **Look** — draw a PCoA map of the samples, coloured by arm. A *PCoA* (principal coordinate analysis) turns the pairwise distances between samples into a 2-D map: points that are close together have similar communities. We use *Bray-Curtis* distance, computed on relative abundances: 0 means two samples share the same species proportions, 1 means nothing in common.
2. **Test** — PERMANOVA (Anderson 2001, https://doi.org/10.1111/j.1442-9993.2001.01070.pp.x). In plain words: it asks "how much of the difference between samples is explained by arm, and is that more than we'd see if we shuffled the arm labels at random?" The R^2 is the "how much" (like R^2 in a regression); the *p-value* comes from comparing it against 999 shuffles of the labels.
3. **Worry** — check that the answer survives different distances, a stricter low-depth filter, and the fact that BV status (which dominates these communities) is tangled up with arm.

Study B has one sample per participant, so there are no repeat visits to worry about.

```
library(tidyverse)
library(here)
library(microViz)     # the pipeline: tax_transform |> dist_calc |> ord_calc |> ord_plot
library(phyloseq)     # prune_samples() and friends come from here
library(patchwork)    # lets us put two ggplots side by side with p1 + p2

source(here("R/count_tables.R"))

d  <- read_counts("group_b")                        # plain tables (metadata, species counts
ps <- as_counts("group_b", "microbiome", data = d)  # same data as a phyloseq object for mic
```

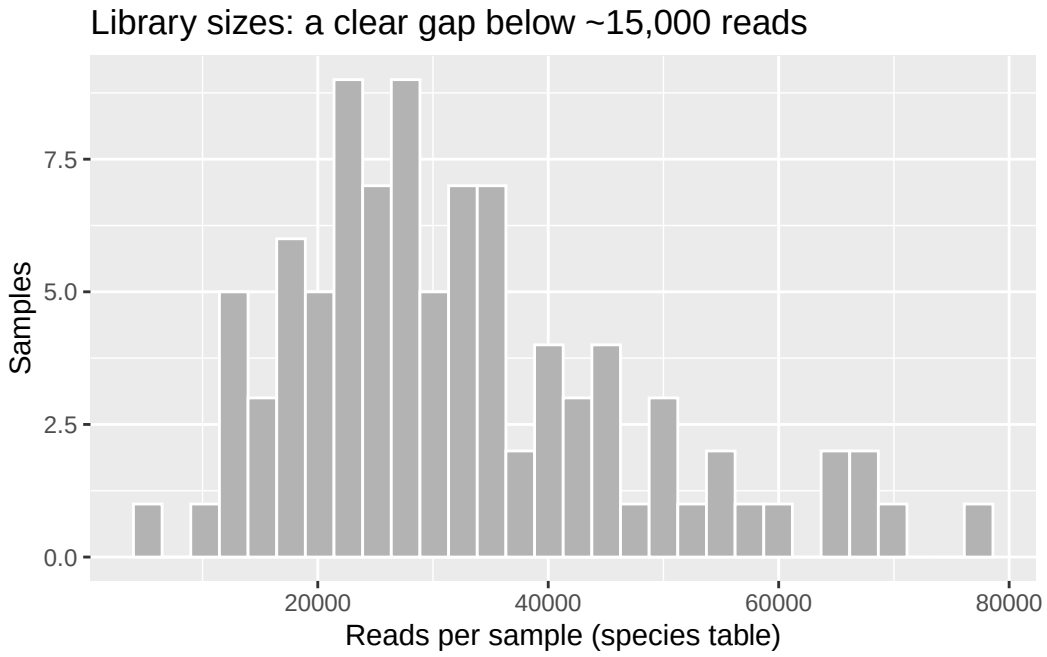
```
# The working object for most of the report: Bray-Curtis distance on
# RELATIVE abundances. "compositional" turns each sample's counts into
# shares of that sample's total first, so the distance compares proportions.
# (On raw counts, a shallow sample looks different just for being shallow,
# and we have nine shallow samples - see section 0.)
psb <- ps |> tax_transform("compositional") |> dist_calc("bray")
```

0. A look at depth first

Bray-Curtis uses *proportions*, so shallow samples are noisier: with few reads, the proportions are estimated badly. Before anything else, where do the library sizes sit?

```
lib <- tibble(sample_id = d$metadata$sample_id,
              arm       = d$metadata$arm,
              reads     = colSums(d$species))

ggplot(lib, aes(reads)) +
  geom_histogram(bins = 30, fill = "grey70", colour = "white") +
  labs(x = "Reads per sample (species table)", y = "Samples",
       title = "Library sizes: a clear gap below ~15,000 reads")
```



There is a natural break in the histogram: nine samples sit at 14,500 reads or below and the next one up is at 15,200. I take anything below 15,000 as "low depth" and keep it in mind for the robustness checks later. I do **not** drop these samples from the main analysis — a rule picked by looking at the outcome of the test would be a rule picked to get an answer.

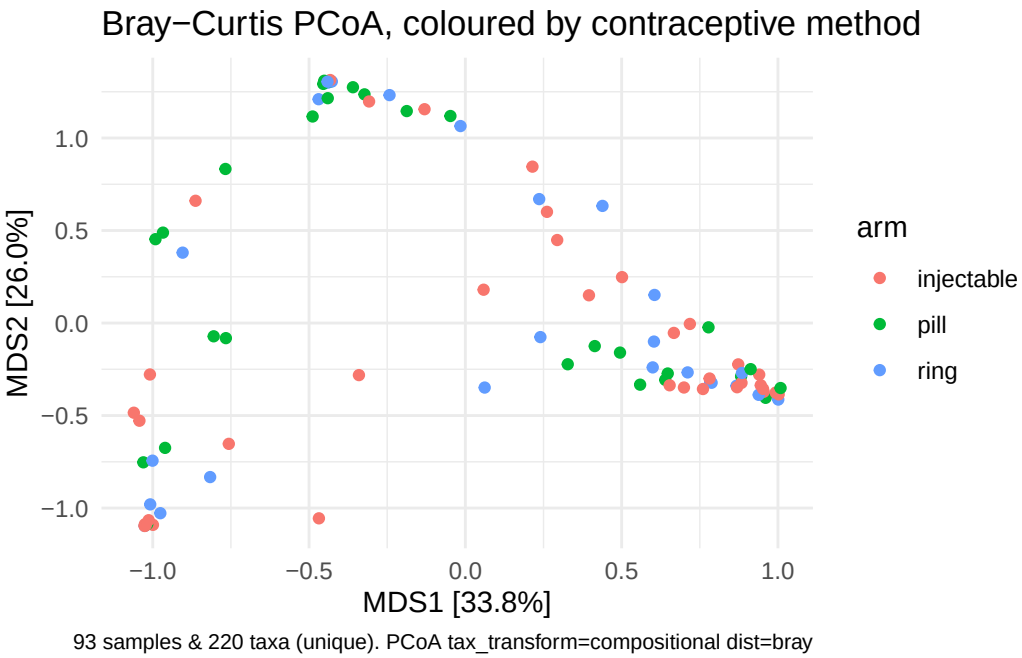
```
# which samples the 15,000 rule catches
lib |> filter(reads < 15000) |> arrange(reads)
```

```
# A tibble: 9 x 3
  sample_id arm        reads
  <chr>     <chr>      <dbl>
1 B006      ring        5247
2 B083      ring       10756
3 B065      injectable 11570
4 B024      injectable 11627
5 B069      ring       12927
6 B070      injectable 13333
7 B091      pill       13543
8 B003      injectable 14057
9 B076      injectable 14534
```

- **Check me:** rerun the histogram chunk; the bar of small samples at the left edge should hold `nrow(filter(lib, reads < 15000))` = 9 samples.

1. Look: the ordination

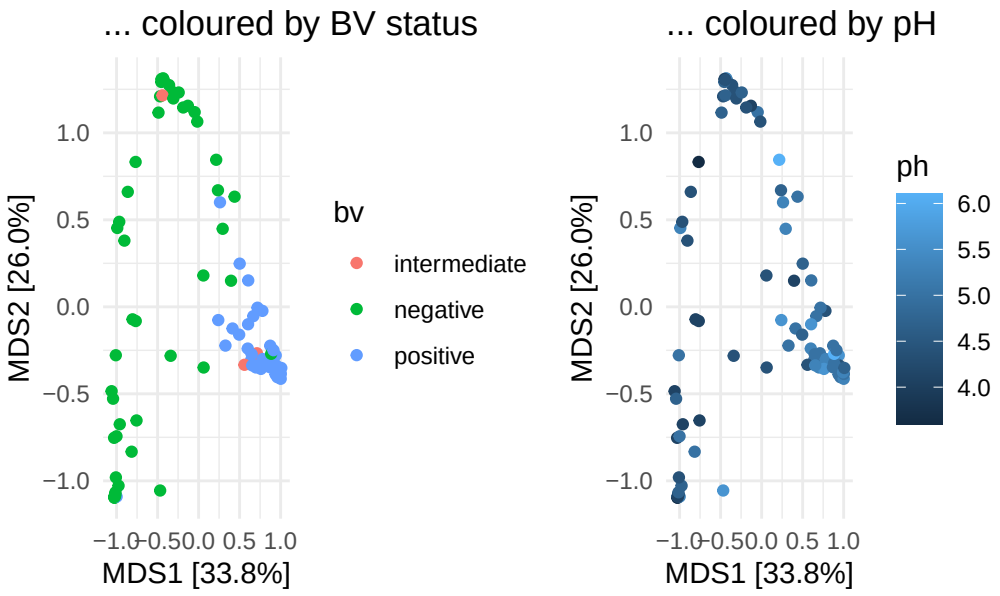
```
# ord_calc("PCoA") builds the map; ord_plot() draws it.
# The axis titles show how much of the total variation each axis carries.
psb |> ord_calc("PCoA") |> ord_plot(colour = "arm") +
  labs(title = "Bray-Curtis PCoA, coloured by contraceptive method")
```



- **Check me:** the plot should have 93 points — `nrow(d$metadata)`. Rerunning the chunk gives exactly the same map: PCoA is deterministic.
- Do the three colours separate into three islands, or overlap? *Your answer:*

The same points coloured by things that might explain more than arm:

```
p1 <- psb |> ord_calc("PCoA") |> ord_plot(colour = "bv") +
  labs(title = "… coloured by BV status")
p2 <- psb |> ord_calc("PCoA") |> ord_plot(colour = "ph") +
  labs(title = "… coloured by pH")
p1 + p2
```



ansform=compositional dist=bray93 samples × 220 taxa (unique). PCoA tax_transform=compositional dist=bray

2. Test: PERMANOVA

Arm on its own, then arm alongside the other sample characteristics. `by = "margin"` means each variable's R^2 is what is left after the others have had their share — without it, the result would depend on the order the variables are listed in. Seed 202 fixes the shuffle order so the numbers are reproducible.

```
# arm alone
psb |> dist_permanova(variables = "arm", n_perms = 999, seed = 202)
```

```
psExtra object - a phyloseq object with extra slots:

phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 220 taxa and 93 samples ]
sample_data() Sample Data:       [ 93 samples by 8 sample variables ]
tax_table()   Taxonomy Table:    [ 220 taxa by 8 taxonomic ranks ]

otu_get(counts = TRUE)       [ 220 taxa and 93 samples ]

psExtra info:
tax_agg = "unique" tax_trans = "compositional"
```

```
bray distance matrix of size 93
0.6550832 0.7350117 0.2355842 0.8155489 0.90071 ...

permanova:
Permutation test for adonis under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 999

vegan::adonis2(formula = formula, data = metadata, permutations = n_perms, by = by, parallel
         Df SumOfSqs      R2      F Pr(>F)
arm       2   0.9504 0.03261 1.5171  0.147
Residual 90  28.1894 0.96739
Total    92  29.1398 1.00000
```

```
# arm together with the other characteristics, including sequencing batch
psb |> dist_permanova(variables = c("arm", "bv", "ph", "any_sti", "rnaseq_batch"),
                      n_perms = 999, seed = 202, by = "margin")
```

```
psExtra object - a phyloseq object with extra slots:

phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 220 taxa and 93 samples ]
sample_data() Sample Data:       [ 93 samples by 8 sample variables ]
tax_table()   Taxonomy Table:    [ 220 taxa by 8 taxonomic ranks ]

otu_get(counts = TRUE)       [ 220 taxa and 93 samples ]

psExtra info:
tax_agg = "unique" tax_trans = "compositional"

bray distance matrix of size 93
0.6550832 0.7350117 0.2355842 0.8155489 0.90071 ...

permanova:
Permutation test for adonis under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 999

vegan::adonis2(formula = formula, data = metadata, permutations = n_perms, by = by, parallel
```

```
             Df SumOfSqs      R2      F Pr(>F)
arm           2   0.7887 0.02706 1.6720  0.108
bv            2   4.1794 0.14343 8.8607  0.001 ***
ph            1   0.3207 0.01101 1.3600  0.223
any_sti       1   0.2143 0.00735 0.9087  0.442
rnaseq_batch  1   0.1640 0.00563 0.6953  0.606
Residual     85  20.0463 0.68793
Total        92  29.1398 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- **Check me:** change the seed to 456 and rerun the first PERMANOVA chunk. The R^2 should stay around 3% and the p-value near 0.15. If the verdict flips when the seed changes, it was never a solid verdict.

2½. A PERMANOVA caveat: different places, or just more spread out?

PERMANOVA can come out significant for a boring reason: one group being more *scattered* than the others, rather than sitting in a different place. `betadisper` (Anderson 2006, https://doi.org/10.1139/E06-036) tests exactly that — "do the groups differ in how spread out they are?" — and a large p-value here is good news: it means the PERMANOVA above is about *where* the groups sit, not how scattered they are. (The warning about "negative squared distances" that this chunk prints under the hood is expected — it comes from the PCoA step inside `betadisper` and does not affect the test.)

```
# dist_get() pulls the distance matrix back out of psb.
dist_b <- dist_get(psb)

set.seed(202)
vegan::permutest(vegan::betadisper(dist_b, d$metadata$bv),  permutations = 999)$tab
```

```
          Df     Sum Sq    Mean Sq         F N.Perm Pr(>F)
Groups     2 0.03313067 0.01656534 0.6578619    999  0.517
Residuals 90 2.26625113 0.02518057        NA     NA     NA
```

```
set.seed(202)
vegan::permutest(vegan::betadisper(dist_b, d$metadata$arm), permutations = 999)$tab
```

```
          Df      Sum Sq     Mean Sq         F N.Perm Pr(>F)
Groups     2 0.008211798 0.004105899 0.2425162    999  0.777
Residuals 90 1.523736784 0.016930409        NA     NA     NA
```

Neither is close to significant (BV: p 0.5; arm: p 0.8): the groups are equally spread out. So the BV signal above — the thing that actually structures these communities — is a genuine shift, not a scatter artefact.

3. Which arms differ?

An overall test of three groups can miss a difference that exists between only two of them, so we test the three pairs separately. Caveat: three tests means three chances to be fooled by chance. The last column applies a **Holm correction**, which pays that back — treat anything that only survives the uncorrected column as *suggestive*, not settled.

```
pairs <- list(c("pill", "injectable"), c("pill", "ring"), c("injectable", "ring"))

pairwise <- map_dfr(pairs, function(pr) {
  # prune_samples() keeps only the listed samples. We find the IDs with an
  # ordinary dplyr filter first, because subset_samples() would need to see
  # `pr` from inside the sample table, where it can't look.
  ids <- d$metadata$sample_id[d$metadata$arm %in% pr]
  t <- as.data.frame(perm_get(
    prune_samples(ids, ps) |>
      tax_transform("compositional") |> dist_calc("bray") |>
      dist_permanova(variables = "arm", n_perms = 999, seed = 202, verbose = FALSE)
  ))
  tibble(comparison = paste(pr[1], "vs", pr[2]),
         r2 = round(t$R2[1], 3), p_value = t$`Pr(>F)`[1])
})

# p_holm: adjusted for having run three tests. Roughly, the smallest
# p-value gets multiplied by 3, the next by 2, the last is left alone.
pairwise |>
  mutate(p_holm = round(p.adjust(p_value, "holm"), 3))
```

```
# A tibble: 3 x 4
  comparison            r2 p_value p_holm
  <chr>              <dbl>   <dbl>  <dbl>
1 pill vs injectable 0.039   0.033  0.099
```

```
2 pill vs ring       0.013   0.602  0.65
3 injectable vs ring 0.017   0.325  0.65
```

4. Worry: robustness

Four checks: the same distance on raw counts, two other distances, and the 15,000-read filter from section 0.

- *Raw-count Bray-Curtis* is what we'd get if we forgot to convert to proportions: shallow samples then look different just for being shallow.
- *Aitchison* (CLR transform + Euclidean distance) treats the data as compositional and works on log-ratios instead of raw proportions.
- *Jaccard* ignores how abundant a species is — only whether it is there.
- Dropping the nine low-depth samples asks whether shallow samples were adding noise that drowned the signal.

```
# A helper that runs the PERMANOVA for arm and returns just its R2 and p.
test_arm <- function(ps_obj) {
  t <- as.data.frame(perm_get(
    ps_obj |> dist_permanova(variables = "arm", n_perms = 999, seed = 202, verbose = FALSE)
  ))
  tibble(r2 = round(t$R2[1], 3), p_value = t$`Pr(>F)`[1])
}

low_depth <- lib$sample_id[lib$reads < 15000]

robust <- bind_rows(
  `Bray (proportions)`       = test_arm(psb),
  `Bray (raw counts)`        = test_arm(ps |> tax_transform("identity") |> dist_calc("bray")),
  `Aitchison (clr + euc)`    = test_arm(ps |> tax_transform("clr", rank = "species") |> dist_
  `Jaccard`                   = test_arm(ps |> tax_transform("binary") |> dist_calc("jaccard"
  `Bray (prop.), low-depth dropped` = test_arm(
    prune_samples(setdiff(d$metadata$sample_id, low_depth), ps) |>
      tax_transform("compositional") |> dist_calc("bray")),
  .id = "approach"
)
robust
```

```
# A tibble: 5 x 3
  approach                          r2 p_value
  <chr>                          <dbl>   <dbl>
```

```
1 Bray (proportions)              0.033   0.147
2 Bray (raw counts)               0.033   0.105
3 Aitchison (clr + euc)           0.028   0.18
4 Jaccard                         0.026   0.19
5 Bray (prop.), low-depth dropped 0.057   0.011
```

5. A sanity check from the biology

If the pill's communities really sit apart, a much simpler summary should show it too: the total share held by *Lactobacillus* species, the genus that dominates healthy vaginal communities.

```
tax <- d$taxonomy
lacto_rows <- tax$Genus == "Lactobacillus"
lacto_share <- colSums(d$species[lacto_rows, ]) / colSums(d$species)

d$metadata |>
  mutate(lacto_share = lacto_share[sample_id]) |>
  group_by(arm) |>
  summarise(median_lacto_share = scales::percent(median(lacto_share), 0.1))
```

```
# A tibble: 3 x 2
  arm        median_lacto_share
  <chr>      <chr>
1 injectable 45.3%
2 pill       84.8%
3 ring       44.7%
```

Verdict

The three methods' communities **overlap far more than they differ**. On the Bray-Curtis map, arm explains about **3%** of the differences between samples (p 0.15) — not significant on its own, and it stays at ~3% (p 0.11) once BV, pH, STI status and sequencing batch have had their share. What clearly *does* structure these communities is **BV status** (R^2 14%, p 0.001), and BV is distributed fairly evenly across the three arms, so it is a rival explanation of the pattern we went looking for rather than a confounder of a real arm effect. The dispersion check in section 2½ confirms that the BV signal is a genuine shift, not an artefact of one group being more scattered.

The pairwise tests point the same way as the sanity check: the **pill** group (median *Lactobacillus* share 85%, vs ~45% for the other two arms) separates from the **injectable** group (p 0.03,

though 0.1 once the three comparisons are corrected for), while the ring sits between and differs from neither. Dropping the nine low-depth samples strengthened the overall arm signal (R^2 6%, p 0.01) — the shallow samples were adding noise — which makes pill-vs-injectable more believable, but it is one comparison out of three and deserves the gene-table follow-up before we call it.

Bottom line for the group: the overall answer is "not much", with one suggestive exception (pill vs injectable) worth chasing tomorrow with the gene table.

Choices made

- Used **Bray-Curtis on relative abundances** (compositional transform) as the main distance; alternatives run as robustness checks: Bray-Curtis on raw counts, Aitchison (CLR + Euclidean) and Jaccard. (The original version of this report used raw counts as the main distance, which the text described as "relative abundances" — they are not the same thing, and raw counts make shallow samples look artificially different.)
- Tested with **PERMANOVA, 999 permutations, seed 202** for reproducibility; an alternative is 9999 permutations, which only steadies the third decimal.
- Used `by = "margin"` in the multi-variable model so each variable's R^2 does not depend on listing order; the alternative (`by = "terms"`) was not used.
- Included **sequencing batch** in the multi-variable model (it was negligible, R^2 0.5%); the alternative was to leave it out.
- Defined **low depth as $<$ 15,000 reads**, read off the gap in the library size histogram, kept those samples in the main analysis and did a **sensitivity analysis** without them; the alternative is to drop them throughout, which moves the headline p-value from ~0.15 to ~0.01.
- Reported **pairwise tests with a Holm correction** alongside the raw p-values; the alternative (raw p-values only, yesterday's version) makes pill-vs-injectable look stronger than it is: 0.03 raw, 0.10 corrected.
- Checked **multivariate dispersion** (`betadisper`) so the PERMANOVA results are not a spread artefact; the alternative was to trust PERMANOVA alone, which can be significant purely because one group is more scattered.
- Asked the question of the **species table only**; the same pipeline runs on the gene table and is the next step.